SIR AKBAR SCIENCE ACADEMY
10th MATHEMATICS (NEW BOOK)
EXERCISE 11.1

1. *The following frequency distribution represents the achievement of a student's group upon a memory test:*

(i) Plot cumulative frequency graph of group scores.

(ii) Determine median, lower and upper quartiles, D_3, P_{90} and interquartile range graphically.

Note: Write your answers in whole numbers.

Scores	44–48	49–53	54–58	59–63	64–68	69–73	74–78
No. of Students (f)	3	12	15	23	12	8	7

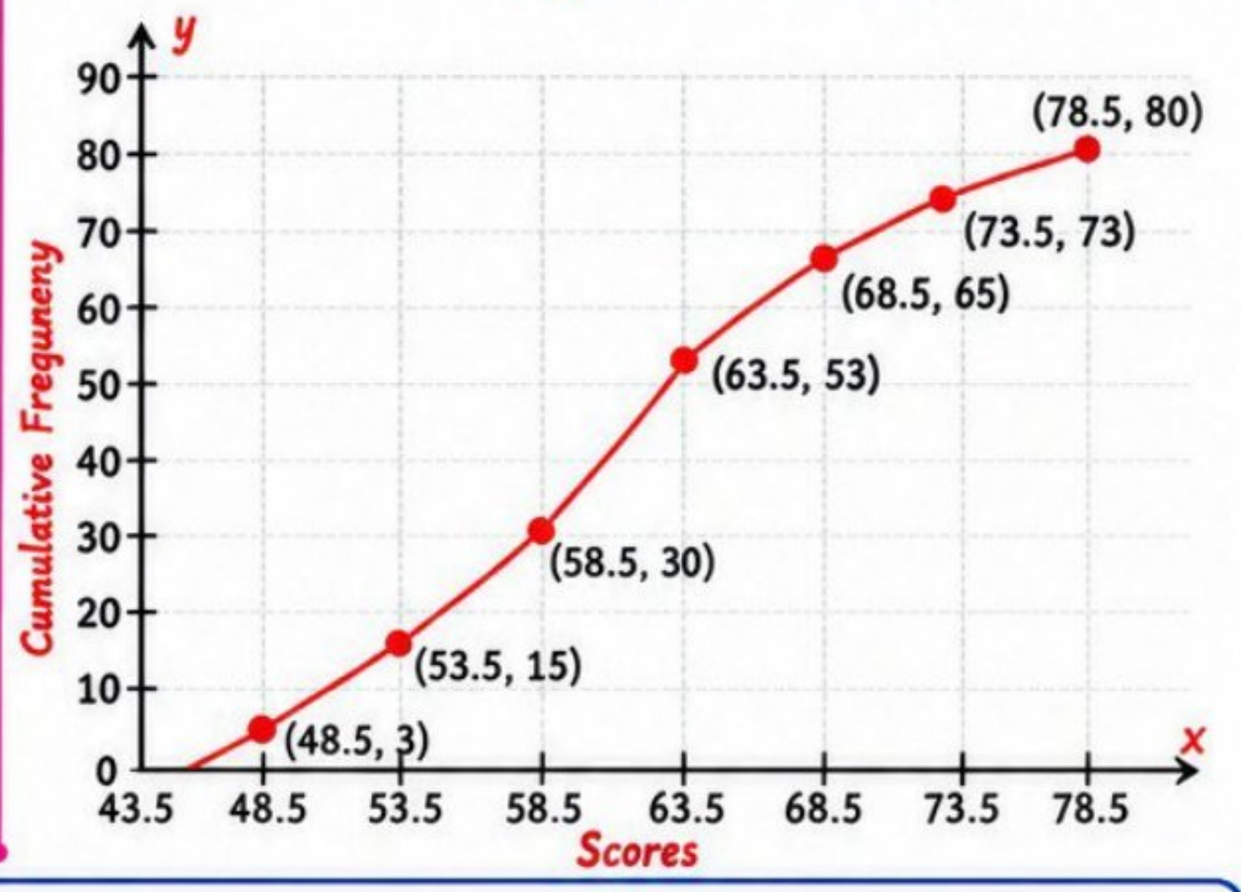
★ STEP 1: Construct Cumulative Frequency Table

Scores	f	Cumulative Frequency (c.f.)
44–48	3	3
49–53	12	15
54–58	15	30
59–63	23	53
64–68	12	65
69–73	8	73
74–78	7	80

Total observations, $n = \sum f = 80$

★ STEP 2: Plot the Less-than Ogive

Plot the points using upper class boundaries.



STEP 3: Find Required Positions

(i) Median

$$\frac{n}{2} = \frac{80}{2} = 40$$

From the graph, at c.f. = 40, the reading on x-axis is 61.

Median = 61

(ii) First Quartile (Q_1)

$$\frac{n}{4} = \frac{80}{4} = 20$$

From the graph, at c.f. = 20, the reading on x-axis is 55.

$Q_1 = 55$

(iii) Third Quartile (Q_3)

$$\frac{3n}{4} = \frac{3(80)}{4} = 60$$

From the graph, at c.f. = 60, the reading on x-axis is 66.

$Q_3 = 66$

(iv) Third Decile (D_3)

$$\frac{3n}{10} = \frac{3(80)}{10} = 24$$

From the graph, at c.f. = 24, the reading on x-axis is 57.

$D_3 = 57$

(v) 90th Percentile (P_{90})

$$\frac{90n}{100} = \frac{90(80)}{100} = 72$$

From the graph, at c.f. = 72, the reading on x-axis is 73.

$P_{90} = 73$

(vi) Interquartile Range (IQR)

$$IQR = Q_3 - Q_1$$
$$= 66 - 55$$

IQR = 11

FINAL ANSWERS

Median = 61 | $Q_1 = 55$ | $Q_3 = 66$ | $D_3 = 57$ | $P_{90} = 73$ | IQR = 11

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10th MATHEMATICS (NEW BOOK)
EXERCISE 11.1

2. Construct an ogive for the following distribution of scores:

Scores	69 – 79	79 – 89	89 – 99	99 – 109	109 – 119
f	4	10	13	8	5

Determine median, lower and upper quartiles, D_6, P_{80} and interquartile range graphically.

Note: Write your answers in whole numbers.

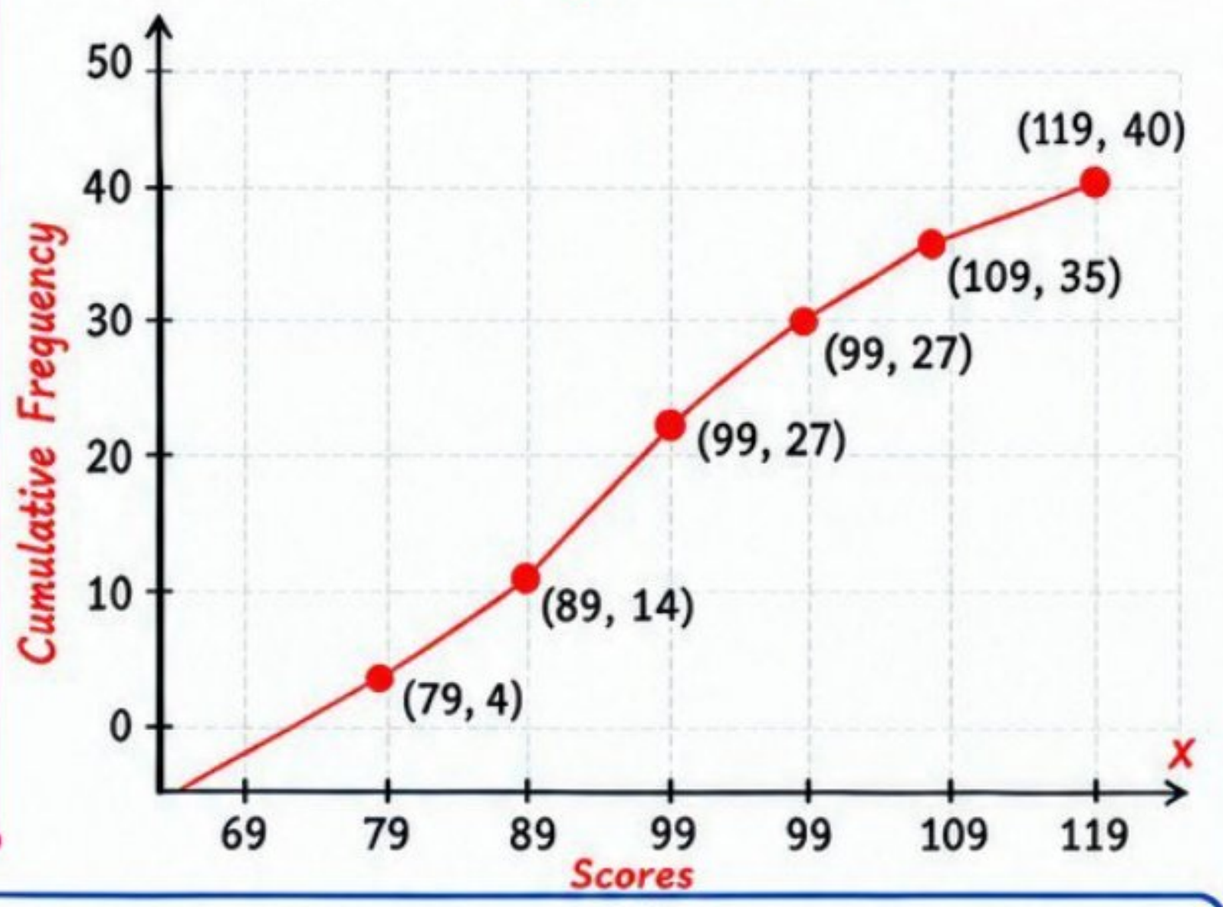
☆ STEP 1: Construct Cumulative Frequency Table

Scores	f	Cumulative Frequency (c.f.)
69 – 79	4	4
79 – 89	10	14
89 – 99	13	27
99 – 109	8	35
109 – 119	5	40

Total observations, $n = \sum f = 40$

☆ STEP 2: Plot the Less-than Ogive

Plot the points using upper class boundaries.



STEP 3: Find Required Positions

(i) Median

$$\frac{n}{2} = \frac{40}{2} = 20$$

From the graph, at c.f. = 40, the reading on x-axis is 61.

Median = 61

(ii) First Quartile (Q_1)

$$\frac{n}{4} = \frac{40}{4} = 20$$

From the graph, at c.f. = 20, the reading on x-axis is 55.

$Q_1 = 55$

(iii) Third Quartile (Q_3)

$$\frac{3n}{4} = \frac{3(80)}{4} = 60$$

From the graph, at c.f. = 60, the reading on x-axis is 66.

$Q_3 = 66$

(iv) Sixth Decile (D_6)

$$\frac{6n}{10} = \frac{6(40)}{10} = 24$$

From the graph, at c.f. = 24, the reading on x-axis is 57.

$D_6 = 57$

(v) 90th Percentile (P_{90})

$$\frac{90n}{100} = \frac{90(80)}{100} = 72$$

From the graph, at c.f. = 72, the reading on x-axis is 73.

$P_{90} = 73$

(vi) Interquartile Range (IQR)

$$IQR = Q_3 - Q_1$$
$$= 66 - 55$$

IQR = 11

FINAL ANSWERS

Median = 61 | $Q_1 = 55$ | $Q_3 = 66$ | $D_3 = 57$ | $P_{90} = 73$ | IQR = 11

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10th MATHEMATICS (NEW BOOK)
EXERCISE 11.1

4. Find Q_1, Q_3, median, range, IQR and extreme values plot for the following data:

102, 98, 95, 100, 93, 110, 108, 104, 97, 96, 92, 101, 99, 105, 107

Also draw a box-and-whisker plot.

STEP 1: Arrange the data in ascending order.

92, 93, 95, 96, 97, 98, 99, 100,
101, 102, 104, 105, 107, 108, 110

Total number of observations, $n = 15$

STEP 2: Find the Median.

$$\text{Position of median} = \frac{n+1}{2} = \frac{15+1}{2} = 8$$

The 8th observations = 100

Median = 100

STEP 3: Find the First Quartile (Q_1).

Lower half (excluding median):

92, 93, 95, **96**, 97, 98, 99

Middle value = Q_1 = 96

STEP 4: Find the Third Quartile (Q_3).

Upper half (excluding median):

101, 102, 104, **105**, 107, 108, 110

Middle value = Q_3 = 105

STEP 5: Find the Range.

Highest value (H) = 110

Lowest value (L) = 92

Range = H − L = 110 − 92 = 18

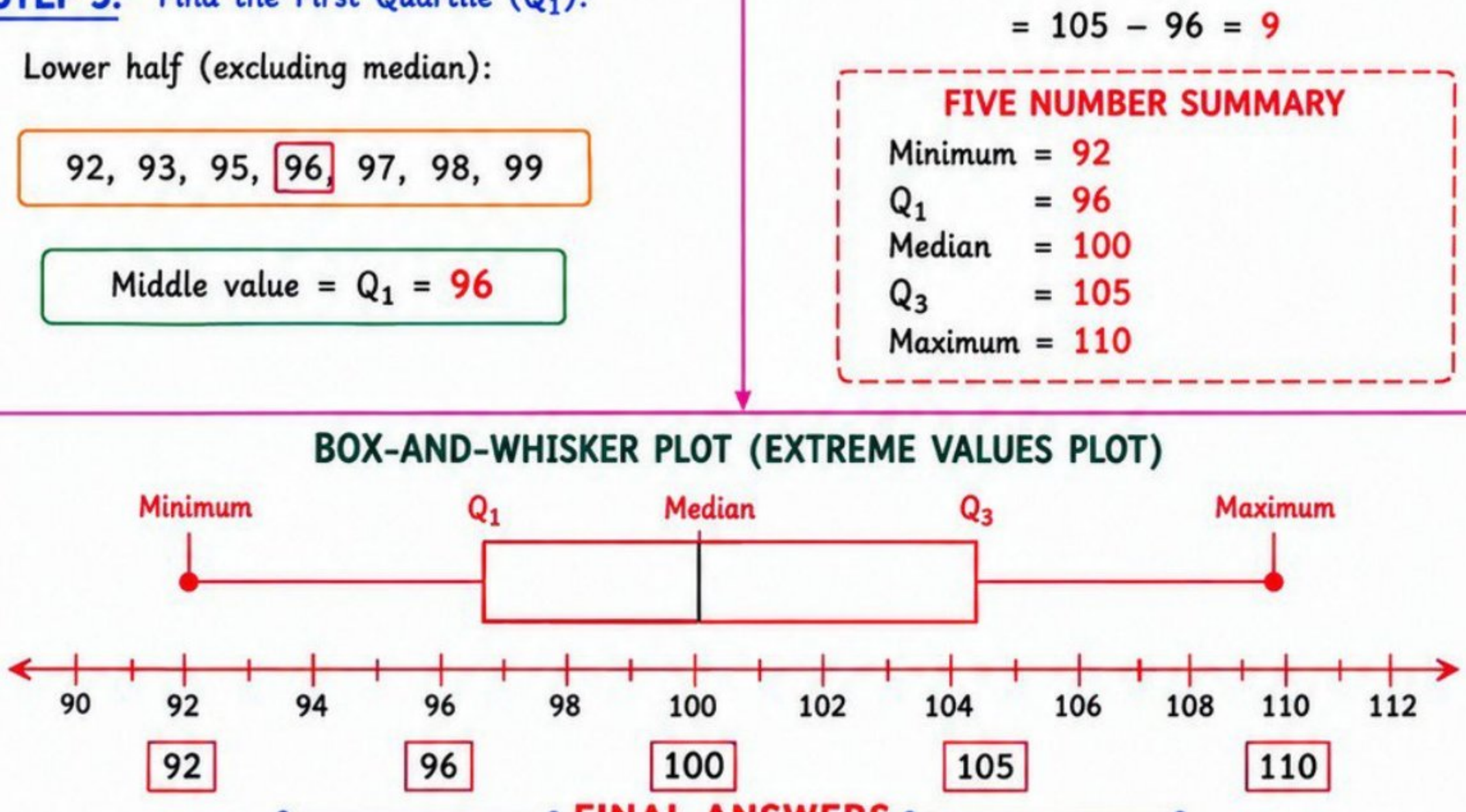
STEP 6: Find the Interquartile Range (IQR).

$$IQR = Q_3 - Q_1$$
$$= 105 - 96 = 9$$

FIVE NUMBER SUMMARY

Minimum = 92
Q_1 = 96
Median = 100
Q_3 = 105
Maximum = 110

BOX-AND-WHISKER PLOT (EXTREME VALUES PLOT)



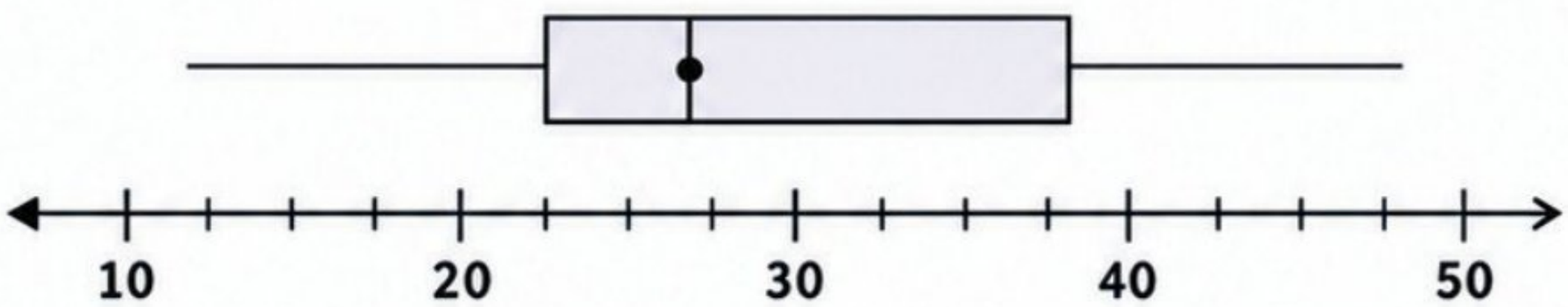
FINAL ANSWERS

Minimum = 92 | Q_1 = 96 | Q_3 = 105 | D_3 = 107 | P_{90} = 103 | IQR = 18

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10th MATHEMATICS (NEW BOOK)
EXERCISE 11.1

5. Find Q_1, Q_3, median, minimum and maximum values for the following box-and-whisker plot:



Note:
Read the values directly from the box-and-whisker plot.

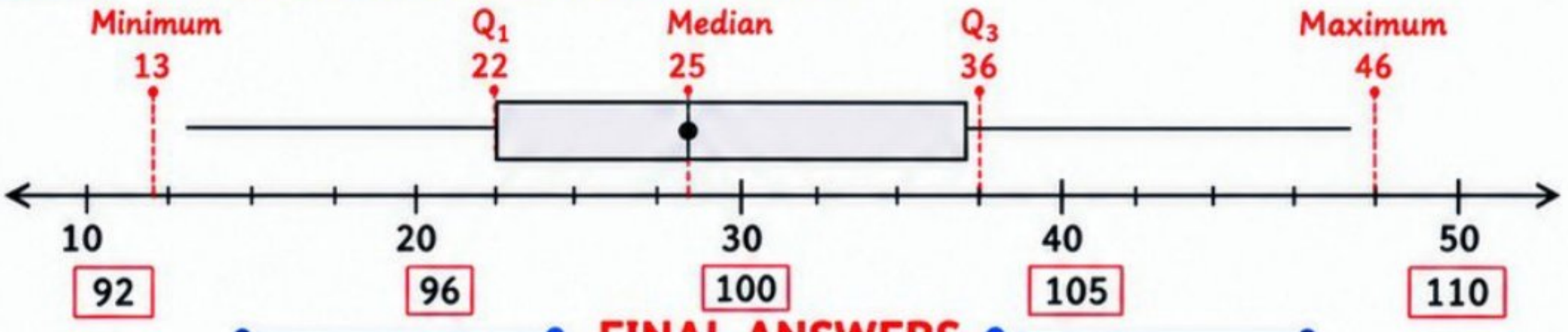
☆ STEP 1: Understand the parts of box-and-whisker plot.

- Left end of the left whisker = Minimum value
- Left edge of the box = First quartile (Q_1)
- Line (or dot) inside the box = Median
- Right edge of the box = Third quartile (Q_3)
- Right end of the right whisker = Maximum value

☆ STEP 2: Read the values from the given box-and-whisker plot.

- Minimum value = Left end of left whisker = 13
- Q_1 (First quartile) = Left edge of the box = 22
- Median value = Line (or dot) inside the box = 25
- Q_3 (Third quartile) = Right edge of the box = 36
- Maximum value = Right end of right whisker = 46

☆ STEP 3: Mark and label the values on the number line.



☆ STEP 4: Find the Third Quartile (Q_3).

Upper half (excluding median):

101, 102, 104, **105**, 107, 108, 110

Middle value = Q_3 = 105

• STEP 5: Find the Range.

Highest value (H) = 110

Lowest value (L) = 92

Range = H − L = 110 − 92 = 18

• STEP 6: Find the Interquartile Range (IQR).

$IQR = Q_3 - Q_1$

$= 105 - 96 = 9$

FINAL ANSWERS

Minimum = 92
Q_1 = 96
Median = 100
Q_3 = 105
Maximum = 110

FINAL ANSWERS

Minimum = 92 | Q_1 = 96 | Q_3 = 103 | D_3 = 97 | P_{90} = 73 | IQR = 11

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10th MATHEMATICS (NEW BOOK)

EXERCISE 11.1

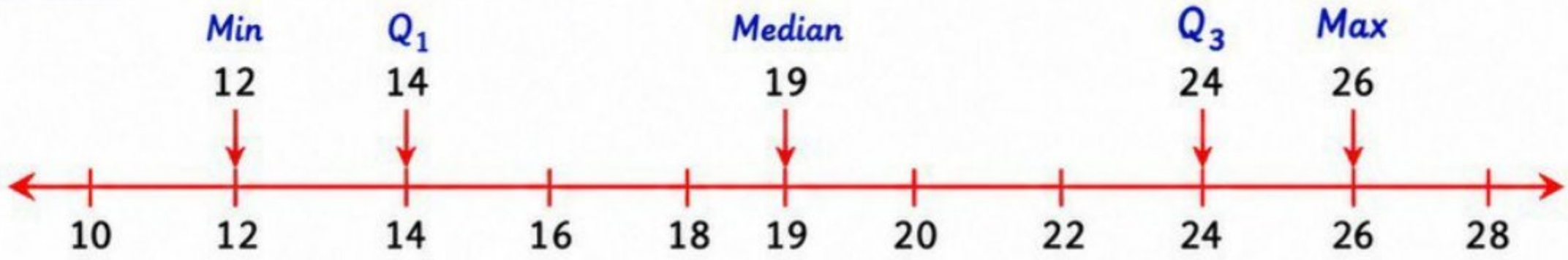
6. Draw the box-and-whisker plot, if min. value = 12, max. value = 26, median = 19, $Q_1 = 14$ and $Q_3 = 24$.

Note:
A box-and-whisker plot is drawn using the five-number summary.

★ **STEP 1:** *Write the five-number summary.*

Statistic	Value
Minimum	12
First Quartile (Q_1)	14
Median	19
Third Quartile (Q_3)	24
Maximum	26

★ **STEP 2:** *Mark the values on a number line.*

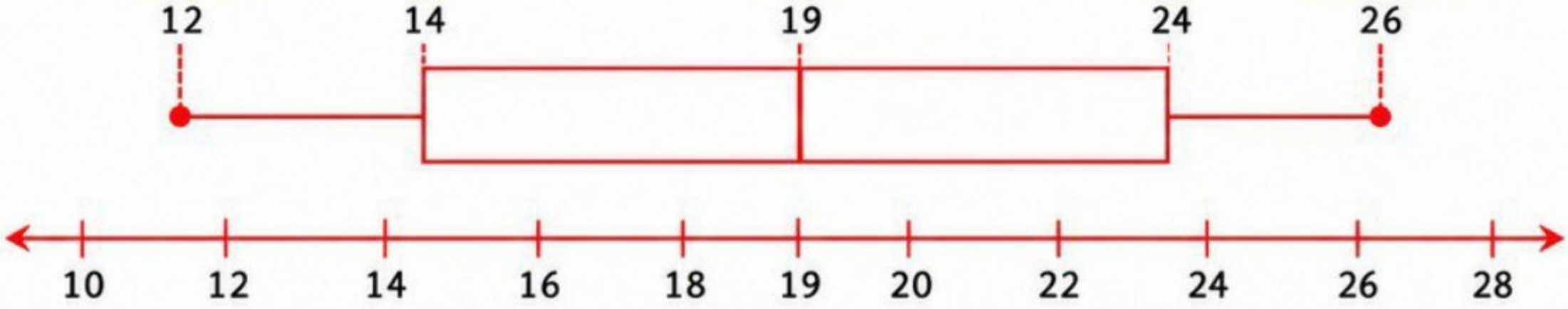


★ **STEP 3:** *Draw the box-and-whisker plot.*

- Left whisker is drawn from Minimum (12) to Q_1 (14).
- Box is drawn from Q_1 (14) to Q_3 (24).
- A line is drawn inside the box at the Median (19).
- Right whisker is drawn from Q_3 (24) to Maximum (26).

FIVE-NUMBER SUMMARY

- Minimum = 12
- Q_1 (First Quartile) = 14
- Median = 19
- Q_3 (Third Quartile) = 24
- Maximum = 26



Therefore, the required box-and-whisker plot is shown above.

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10th MATHEMATICS (NEW BOOK)

EXERCISE 11.1

7. Construct a scatter diagram and draw a line of best fit for the following quiz scores for 8 students in a class:

1st quiz	14	28	34	36	43	45	46	49
2nd quiz	16	20	33	35	38	39	42	43

Describe correlation between 1st and 2nd quiz scores also.

STEP 1: Make a table of ordered pairs and plot the points.

Student	1st Quiz (x)	2nd Quiz (y)	Ordered Pair (x, y)
1	14	16	(14, 16)
2	28	20	(28, 20)
3	34	33	(34, 33)
4	36	35	(36, 35)
5	43	38	(43, 38)
6	43	39	(43, 39)
7	46	42	(45, 42)
8	49	43	(49, 43)

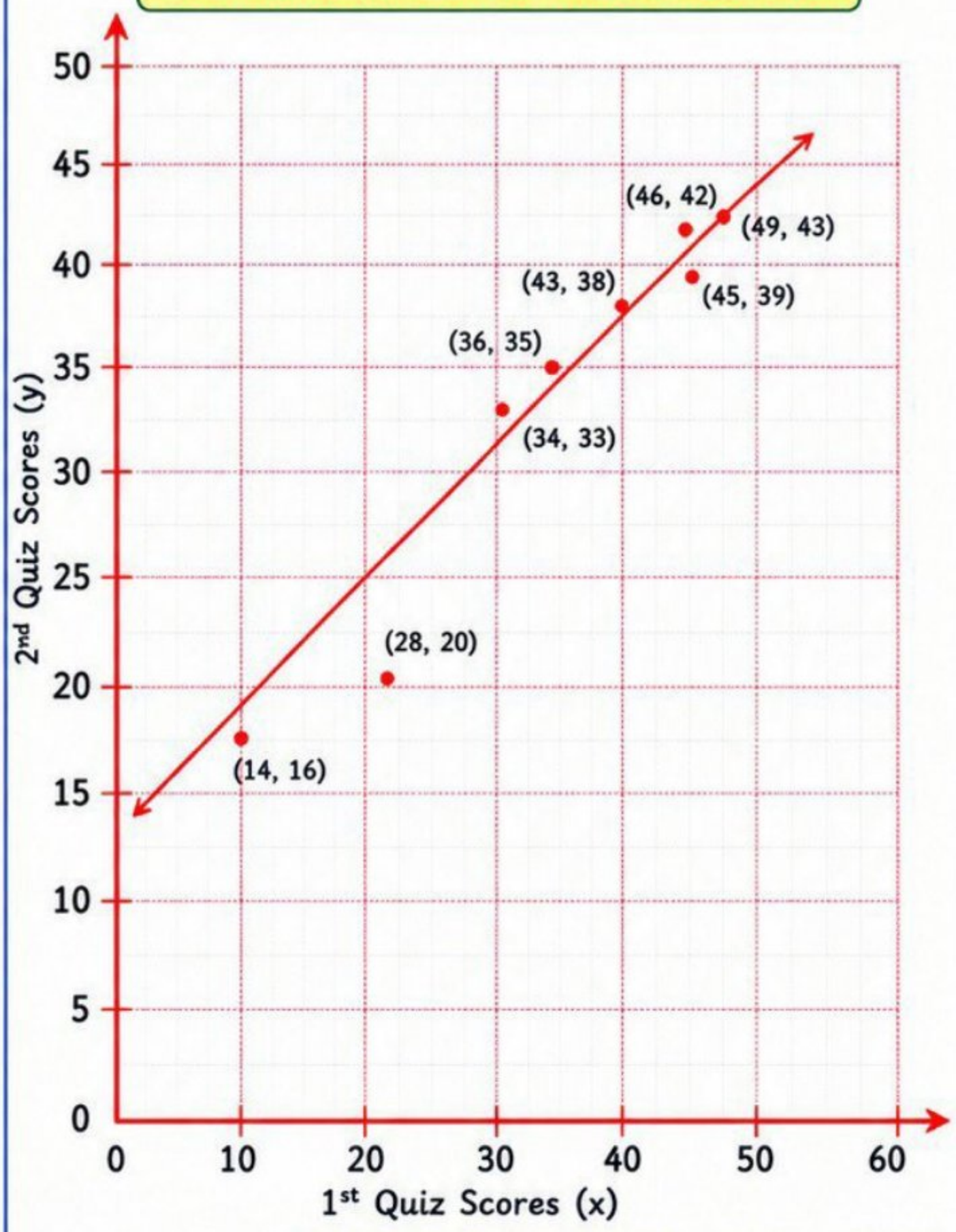
STEP 2: Draw the scatter diagram and a line of best fit.

- Plot all the points from the table on a graph.
- Draw a straight line that passes as close as possible to all the points.

What is a Line of Best Fit?

It is a straight line drawn on a scatter diagram such that the number of points above the line is approximately equal to the number of points below the line and the distances of the points from the line are minimum.

SCATTER DIAGRAM WITH LINE OF BEST FIT



STEP 3: Describe the correlation.

The points are increasing from left to right and close to the line of best fit. Therefore, there is a **STRONG POSITIVE CORRELATION** between 1st and 2nd quiz scores.

Conclusion:

As the 1st quiz score increases, the 2nd quiz score also increases.

Hence, students who perform well in the 1st quiz tend to perform well in the 2nd quiz.

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10th MATHEMATICS (NEW BOOK)

EXERCISE 11.1

8. *The following table shows the test scores (one measuring speed and the other measuring strength) of football players out of 200:*

1st Test (Speed)	20	38	70	40	100	110	50	136	150	80
2nd Test (Strength)	160	142	130	120	70	50	140	44	36	100

(i) *Construct a scatter diagram and draw a line of best fit.*

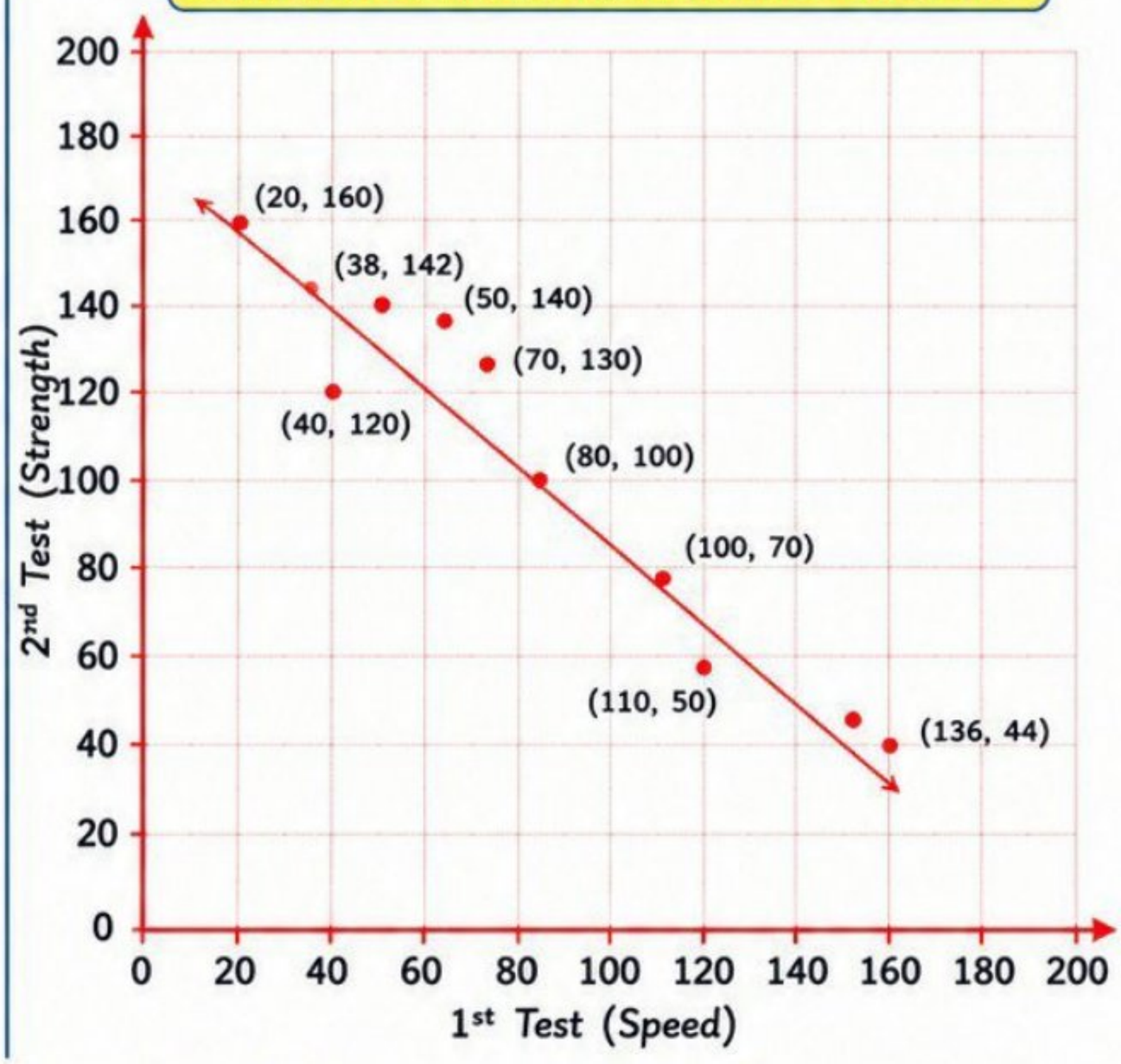
(ii) *Describe correlation between the 1st and 2nd tests.*

(iii) *Abdullah scores 60 in the 2nd test. Estimate his score in the 1st test.*

STEP 1: Make a table of ordered pairs and plot the points.

Player	1st Test (x) Speed	2nd Test (y) Strength	Ordered Pair (x, y)
1	20	160	(20, 160)
2	38	142	(38, 142)
3	70	130	(70, 130)
4	40	120	(40, 120)
5	100	70	(100, 70)
6	110	50	(110, 50)
7	50	140	(50, 140)
8	136	44	(136, 44)
9	150	36	(150, 36)
10	80	100	(80, 100)

SCATTER DIAGRAM WITH LINE OF BEST FIT



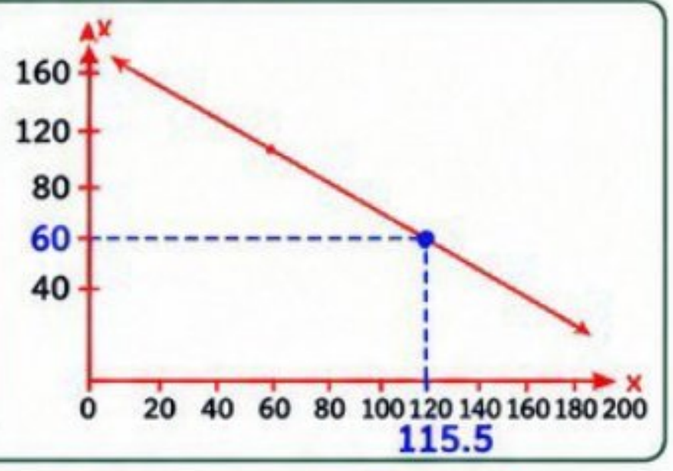
STEP 2: Describe the correlation.

From the scatter diagram, as the 1st Test (Speed) score increases, the 2nd Test (Strength) score decreases.
Therefore, there is a **STRONG NEGATIVE CORRELATION** *between the 1st and 2nd tests.*

Remember!
If the line of best fit slopes downwards from left to right, the correlation is negative. The more the points lie close to the line, the stronger the negative correlation.

STEP 3: Abdullah scores 60 in the 2nd test. Estimate his score in the 1st test.

- *On the graph, take y = 60 on the vertical axis.*
- *Move horizontally to meet the line of best fit.*
- *From the point of intersection, drop a vertical line down to the x-axis.*



Estimated 1st Test (Speed) score ≈ **115.5**

FINAL ANSWERS (i) *Scatter diagram with line of best fit* | (ii) *Strong negative correlation* | (iii) *Estimated 1st Test score* = 115.5

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10th MATHEMATICS (NEW BOOK)

EXERCISE 11.1

9. *The following table shows the number of daily social media hours and GPA of university students.*

Social Media Hours	0.5	1	2	3	4	5	6	7	9
GPA	4.0	3.9	3.7	3.5	3.3	3.1	2.9	2.8	2.5

(i) *Construct a scatter diagram and draw a line of best fit.*

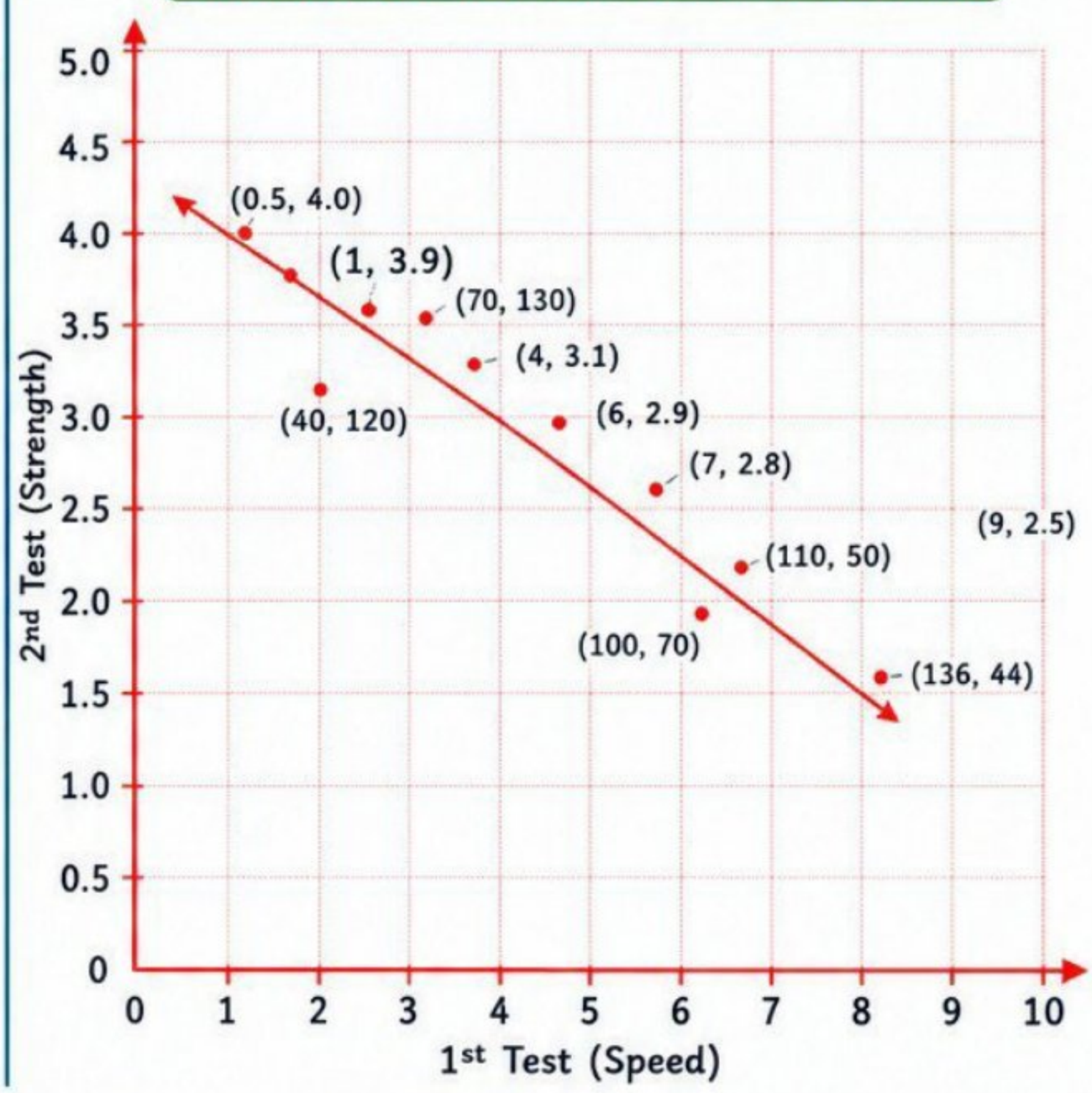
(ii) *Identify the nature of correlation between social media use and GPA.*

(iii) *Estimate the GPA for a student who spends 6.5 hours on social media.*

STEP 1: Make a table of ordered pairs and plot the points.

Student	1st Social Media Hours (x)	2nd Test (y) GPA	Ordered Pair (x, y)
1	0.5	4.0	(0.5, 4.0)
2	1	3.9	(1, 3.9)
3	2	3.7	(2, 3.7)
4	3	3.5	(3, 3.5)
5	4	3.3	(4, 3.3)
6	5	3.1	(5, 3.1)
7	6	2.9	(6, 2.9)
8	7	2.8	(7, 2.8)
9	8	2.6	(8, 2.6)
10	9	2.5	(9, 2.5)

SCATTER DIAGRAM WITH LINE OF BEST FIT



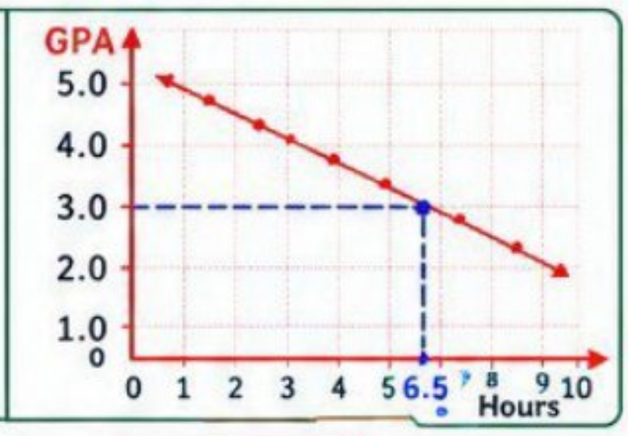
STEP 2: Describe the correlation.

- From the scatter diagram, as the 1st Test (Speed) score increases, the 2nd Test (Strength) score decreases.
- Therefore, there is a **STRONG NEGATIVE CORRELATION** between the 1st and 2nd tests.

Remember!
If the line of best fit slopes downwards from left to right, the correlation is negative. The more the points lie close to the line, the stronger the negative correlation.

STEP 4: Estimate GPA for 6.5 hours.

- *On the graph, take x = 6.5 on the horizontal axis.*
- *Move vertically up to meet the line of best fit.*
- *From the point of intersection, move horizontally left to the vertical axis to read the GPA.*



Estimated 1st Test (Speed) score ≈ 115.5

FINAL ANSWERS

(i) *Scatter diagram with line of best fit* | (ii) *Strong negative correlation* | (iii) *Estimated 1st Test score = 115.5*

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10th MATHEMATICS (NEW BOOK)

EXERCISE 11.2

1. Find the range of the following data sets:

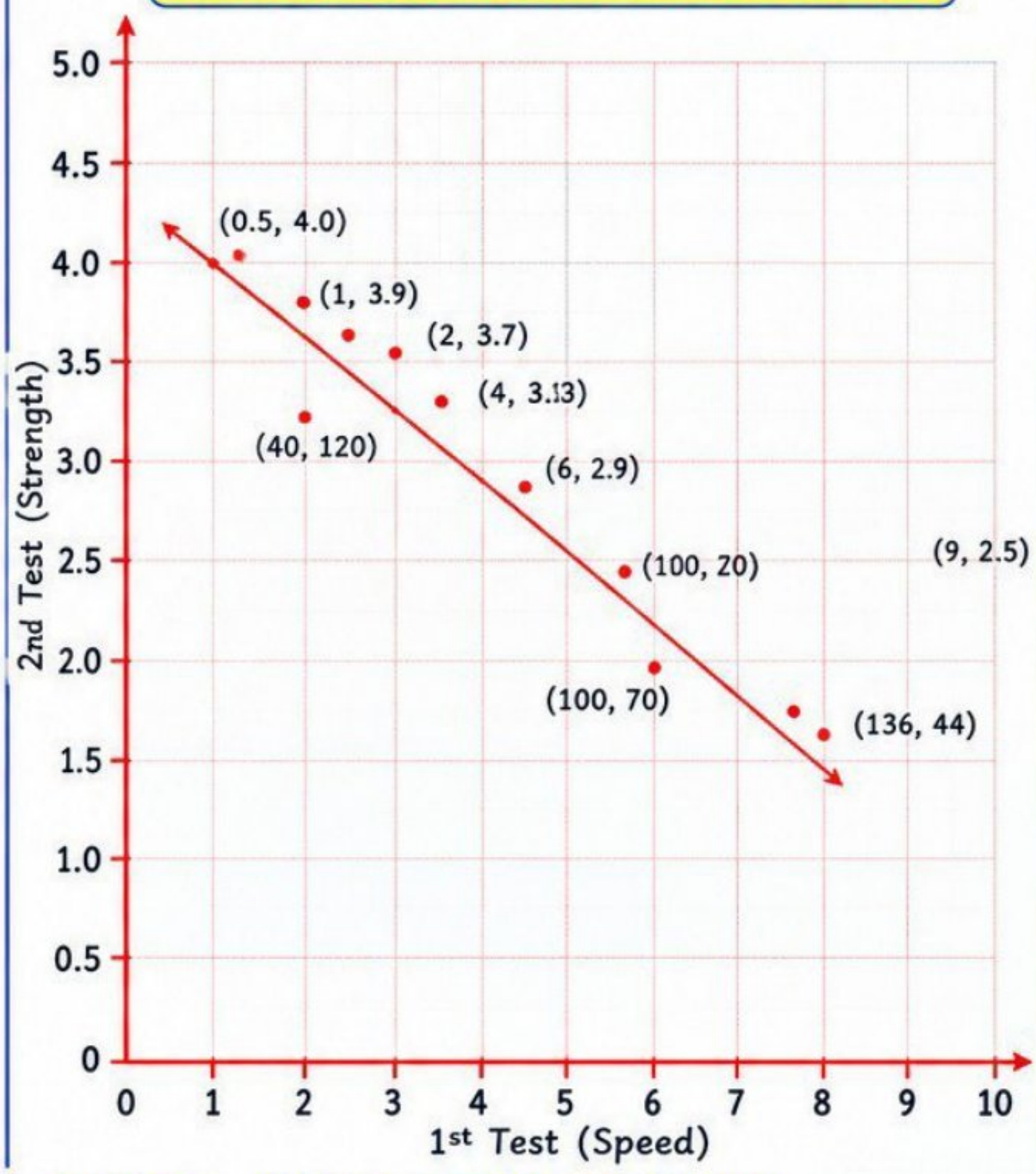
(i) 63, 89, 98, 125, 79, 108, 117, 60 (ii) 43.5, 13.6, 18.9, 38.4, 61.4, 29.4

Remember! Range shows the spread of data. It is the difference between the highest value and the lowest value in the data set.

STEP 1: Make a table of ordered pairs and plot the points.

Student	1st Social Media Hours (x)	2nd Test (y) GPA	Ordered Pair (x, y)
1	0.5	4.0	(0.5, 4.0)
2	1	3.9	(1, 3.9)
3	2	3.7	(2, 3.7)
4	3	3.5	(3, 3.5)
5	4	3.3	(4, 3.3)
6	5	3.1	(5, 3.1)
7	6	2.9	(6, 2.9)
8	7	2.8	(7, 2.8)
9	8	2.6	(8, 2.6)
10	9	2.5	(9, 2.5)

SCATTER DIAGRAM WITH LINE OF BEST FIT



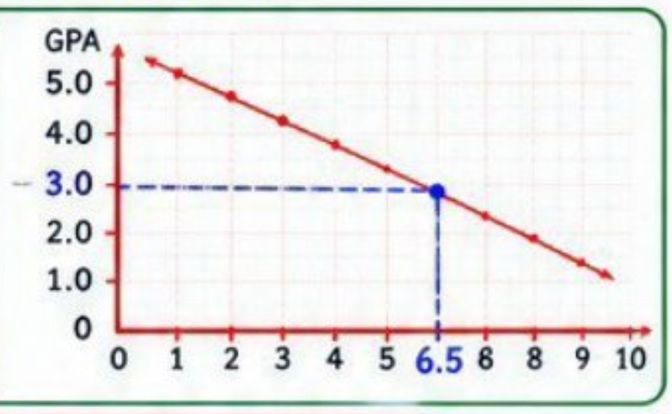
STEP 2: Describe the correlation.

- From the scatter diagram, as the 1st Test (Speed) score increases, the 2nd Test (Strength) score decreases.
- Therefore, there is a STRONG NEGATIVE CORRELATION between the 1st and 2nd tests.

Remember!
If the line of best fit slopes downwards from left to right, the correlation is negative. The more the points lie close to the line, the stronger the negative correlation.

STEP 3: Abdullah scores 60 in the 2nd test. Estimate his score in the 1st test.

- On the graph, take y = 60 on the vertical axis.
- Move horizontally to meet the line of best fit.
- From the point of intersection, drop a vertical line down to the x-axis.



Estimated 1st Test (Speed) score ≈ **115.5**

FINAL ANSWERS

(i) Scatter diagram with line of best fit | (ii) Strong negative correlation | (iii) Estimated 1st Test score = 115.5

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EXERCISE 11.2

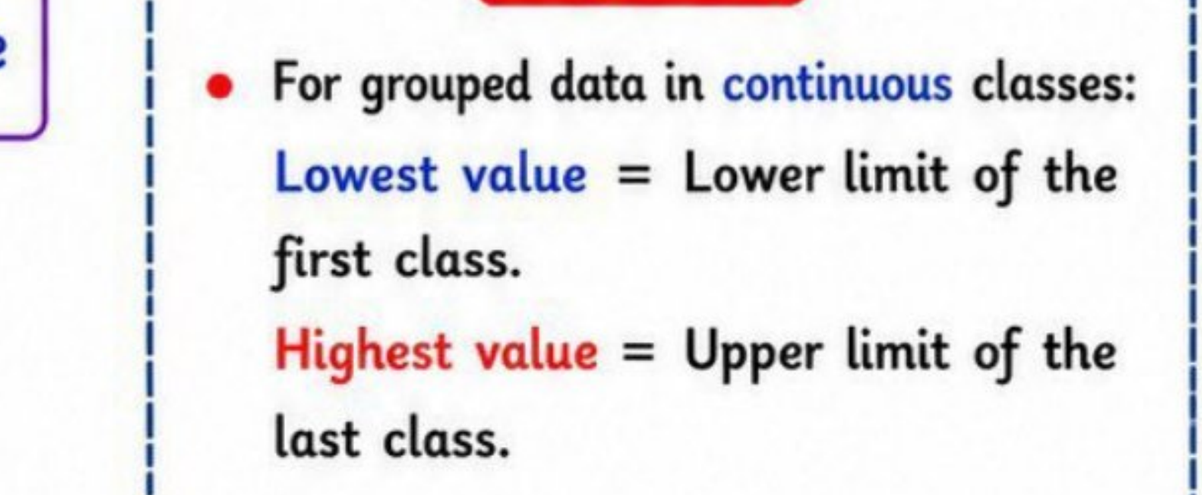
3. Calculate the range of the following data:

Income (in Rs.)	4000 – 4500	4500 – 5000	5000 – 5500	5500 – 6000	6000 – 6500
No. of workers	8	12	30	21	6

Remember! Range shows the spread of data. It is the difference between the highest value and the lowest value in the data set.

STEP 1: Identify the lowest and highest values.

- The lowest value is the lower limit of the first class = 4000
- The highest value is the upper limit of the last class = 6500



STEP 2: Write the formula for range.

Range = Highest Value – Lowest Value

STEP 3: Substitute the values in the formula.

Range = 6500 – 4000

= 2500

Answer: 2500

Key Point

- For grouped data in continuous classes:
  Lowest value = Lower limit of the first class.
  Highest value = Upper limit of the last class.
- Range does not depend on frequency (number of workers).

FINAL ANSWER The range of the given data is 2500.

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SIR AKBAR SCIENCE ACADEMY

10th MATHEMATICS (NEW BOOK)

EXERCISE 11.2

4. A group of 7 workers reported the number of items they assembled in a day as:

52, 55, 50, 53, 54, 56, 52

Find the standard deviation and variance of the items assembled.

STEP 1: Find the mean $(\bar{x})$.

$$\bar{x} = \frac{\Sigma x}{n} \qquad \left[\begin{array}{l} \Sigma x = \text{Sum of all observations} \\ n = \text{Number of observations} \end{array}\right]$$

$$\Sigma x = 52 + 55 + 50 + 53 + 54 + 56 + 52 = 372$$

$$n = 7$$

$$\bar{x} = \frac{372}{7} = 53.14$$

FORMULA USED:

$$\text{Variance } (\sigma^2) = \frac{\Sigma (x - \bar{x})^2}{n}$$

$$\text{Standard Deviation } (\sigma) = \sqrt{\sigma^2}$$

STEP 2: Calculate $(x - \bar{x})$ and $(x - \bar{x})^2$.

x	$x - \bar{x}$ $(= x - 53.14)$	$(x - \bar{x})^2$
52	$52 - 53.14 = -1.14$	$(-1.14)^2 = 1.31$
55	$55 - 53.14 = 1.86$	$(1.86)^2 = 3.45$
50	$50 - 53.14 = -3.14$	$(-3.14)^2 = 9.88$
53	$53 - 53.14 = -0.14$	$(-0.14)^2 = 0.02$
54	$54 - 53.14 = 0.86$	$(0.86)^2 = 0.73$
56	$56 - 53.14 = 2.86$	$(2.86)^2 = 8.16$
52	$52 - 53.14 = -1.14$	$(-1.14)^2 = 1.31$
Total (Σ)		$\Sigma (x - \bar{x})^2 = 24.86$

STEP 3: Find the variance (σ^2).

$$\sigma^2 = \frac{\Sigma (x - \bar{x})^2}{n}$$

$$\sigma^2 = \frac{24.86}{7}$$

$$\sigma^2 = 3.55$$

Variance tells us how much the data values are spread out from the mean.

STEP 4: Find the standard deviation (σ).

$$\sigma = \sqrt{\sigma^2}$$

$$\sigma = \sqrt{3.55}$$

$$\sigma = 1.88$$

Standard deviation is the square root of variance.

FINAL ANSWER:

Variance $(\sigma^2) = 3.55$ | Standard Deviation $(\sigma) = 1.88$

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EXERCISE 11.2

5. A librarian recorded the number of visitors during 5 days of a week.

120, 135, 130, 125, 140

Calculate the variance and standard deviation of visitors.

FORMULAS USED :

$$\text{Variance } (\sigma^2) = \frac{\Sigma(x-\bar{x})^2}{n}$$

$$\text{Standard Deviation } (\sigma) = \sqrt{\sigma^2}$$

STEP 1: Find the mean $(\bar{x})$.

$$\bar{x} = \frac{\Sigma x}{n} \qquad \left[\begin{array}{l} \Sigma x = \text{Sum of all observations} \\ n = \text{Number of observations} \end{array}\right]$$

$$\Sigma x = 120 + 135 + 130 + 125 + 140 = 650$$

$$n = 5$$

$$\bar{x} = \frac{650}{5} = 130$$

Note : Mean is the average value of all the observations.

STEP 2: Prepare a table to find $(x-\bar{x})$ and $(x-\bar{x})^2$.

x	$x-\bar{x}$ $(= x - 130)$	$(x-\bar{x})^2$
120	$120 - 130 = -10$	$(-10)^2 = 100$
135	$135 - 130 = 5$	$(5)^2 = 25$
130	$130 - 130 = 0$	$(0)^2 = 0$
125	$125 - 130 = -5$	$(-5)^2 = 25$
140	$140 - 130 = 10$	$(10)^2 = 100$
Total (Σ)		$\Sigma(x-\bar{x})^2 = 250$

Tip : When we square a negative number, the result is always positive.

STEP 3: Find the variance (σ^2).

$$\sigma^2 = \frac{\Sigma(x-\bar{x})^2}{n}$$

$$= \frac{250}{5}$$

$$\sigma^2 = 50$$

Note : Variance tells us how much the data values are spread out from the mean.

STEP 4: Find the standard deviation (σ).

$$\sigma = \sqrt{\sigma^2}$$

$$= \sqrt{50}$$

$$\sigma = 7.07$$

Note : Standard deviation is the square root of variance.

FINAL ANSWER Variance $(\sigma^2) = 50$ | Standard Deviation $(\sigma) = 7.07$

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10th MATHEMATICS (NEW BOOK)

EXERCISE 11.2

6. Find the range, variance and standard deviation of first 23 odd numbers.

STEP 1: Write the data.

First 23 odd numbers are:

1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45

FORMULAS USED :

Range = Highest value − Lowest value

$$\text{Variance } (\sigma^2) = \frac{\Sigma (x - \bar{x})^2}{n}$$

$$\text{Standard Deviation } (\sigma) = \sqrt{\sigma^2}$$

STEP 2: Find the range.

$$\begin{aligned} \text{Range} &= \text{Maximum} - \text{Minimum} \\ &= 45 - 1 \\ &= \underline{\underline{44}} \end{aligned}$$

Smallest odd number = 1
Largest odd number = 45

STEP 3: Find the mean $(\bar{x})$.

The numbers form an arithmetic sequence.

$$\text{Mean of an A.P.} = \frac{\text{First term} + \text{Last term}}{2}$$

$$\bar{x} = \frac{1 + 45}{2} = \frac{46}{2} = \underline{\underline{23}}$$

Note : When the data is in A.P., mean is the average of first and last term.

STEP 4: Find $\Sigma (x - \bar{x})^2$.

x	$x - \bar{x}$ $(= x - 23)$	$(x - \bar{x})^2$
1	1 − 23 = −22	484
3	3 − 23 = −20	400
5	5 − 23 = −18	324
7	7 − 23 = −16	256
9	9 − 23 = −14	196
11	11 − 23 = −12	144
13	13 − 23 = −10	100
15	15 − 23 = − 8	64
17	17 − 23 = − 6	36
19	19 − 23 = − 4	16
21	21 − 23 = − 2	4
23	23 − 23 = 0	0
Total numbers (n) = 23		

x	$x - \bar{x}$ $(= x - 23)$	$(x - \bar{x})^2$
25	25 − 23 = 2	4
27	27 − 23 = 4	16
29	29 − 23 = 6	36
31	31 − 23 = 8	64
33	33 − 23 = 10	100
35	35 − 23 = 12	144
37	37 − 23 = 14	196
39	39 − 23 = 16	256
41	41 − 23 = 18	324
43	43 − 23 = 20	400
45	45 − 23 = 22	484
$\Sigma (x - \bar{x})^2 = 4048$		

The squares of deviations are symmetrical about the mean. ☺

STEP 5: Find the variance (σ^2).

$$\sigma^2 = \frac{\Sigma (x - \bar{x})^2}{n} = \frac{4048}{23}$$

$\sigma^2 = 176$

Note : Variance tells us how much the data values are spread out from the mean.

STEP 6: Find the standard deviation (σ).

$$\sigma = \sqrt{\sigma^2} = \sqrt{176} = \boxed{13.27}$$

Note : Standard deviation is the square root of variance.

FINAL ANSWER Range = 44 | Variance (σ^2) = 176 | Standard Deviation (σ) = 13.27

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10th MATHEMATICS (NEW BOOK)

EXERCISE 11.2

7. The rainfall recorded in various places of five districts in a week is given below. Find its variance and standard deviation.

Rainfall (in mm) (x)	42	51	54	61	63	71
Number of places (f)	5	13	4	9	5	4

FORMULAS USED :

Mean $(\bar{x}) = \dfrac{\Sigma fx}{\Sigma f}$

Variance $(\sigma^2) = \dfrac{\Sigma f(x-\bar{x})^2}{\Sigma f}$

Standard Deviation $(\sigma) = \sqrt{\sigma^2}$

STEP 1: Make a table and calculate Σf and Σfx.

x	f	fx
42	5	210
51	13	663
54	4	216
61	9	549
63	5	315
71	4	284
	$\Sigma f = 40$	$\Sigma fx = 2237$

Tip : Always find Σf and Σfx first.

STEP 2: Find the mean $(\bar{x})$.

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{2237}{40} = \underline{55.925}$$

Mean represents the average rainfall.

STEP 3: Calculate $f(x - \bar{x})^2$.

x	f	$x - \bar{x})$ $(= x - 55.925)$	$(x-\bar{x})^2$	$f(x-\bar{x})^2$
42	5	42 − 55.925 = -13.925	193.926	5 × 193.926 = 969.630
51	13	51 − 55.925 = -4.925	24.255	13 × 24.255 = 315.315
54	4	54 − 55.925 = -1.925	3.706	4 × 3.706 = 14.824
61	9	61 − 55.925 = 5.075	25.757	9 × 25.757 = 231.815
63	5	63 − 55.925 = 7.075	50.041	5 × 50.041 = 250.205
71	4	71 − 55.925 = 15.075	227.270	4 × 227.270 = 909.080
$\Sigma f = 40$			$\Sigma f(x-\bar{x})^2 = 2690.775$	

We multiply the squared deviation by its frequency. ☺

STEP 4: Find the variance (σ^2).

$$\sigma^2 = \frac{\Sigma f(x-\bar{x})^2}{\Sigma f} = \frac{2690.775}{40} = \boxed{67.27}$$

Note : Variance tells us how much the data values are spread out from the mean.

STEP 5: Find the standard deviation (σ).

$$\sigma = \sqrt{\sigma^2} = \sqrt{67.27} = \boxed{8.20}$$

Note : Standard deviation is the square root of variance.

FINAL ANSWER : Variance $(\sigma^2) = 67.27$ | Standard Deviation $(\sigma) = 8.20$

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10th MATHEMATICS (NEW BOOK)

EXERCISE 11.2

8. Machine A Output (units): 98, 100, 102, 101, 99
Machine B Output (units): 95, 100, 105, 90, 110

(i) Which machine has better performance?
(ii) Which machine is more consistent?

FORMULAS USED:

$$\text{Mean } (\bar{x}) = \frac{\Sigma x}{n}$$

$$\text{Variance } (\sigma^2) = \frac{\Sigma (x - \bar{x})^2}{n}$$

$$\text{Standard Deviation } (\sigma) = \sqrt{\sigma^2}$$

PART A : MACHINE A

STEP 1: Find the mean $(\bar{x})$.

$$\bar{x} = \frac{98 + 100 + 102 + 101 + 99}{5}$$

$$= \frac{500}{5} = \underline{100}$$

STEP 2: Prepare the table.

x	$x - \bar{x}$ $(x - \bar{x} = 100)$	$(x - \bar{x})^2$
98	98 - 100 = -2	4
100	100 - 100 = 0	0
102	102 - 100 = 2	4
101	101 - 100 = 1	1
99	99 - 100 = -1	1
$\Sigma x = 500$		$\Sigma (x - \bar{x})^2 = 10$

STEP 3: Find the variance (σ^2).

$$\sigma^2 = \frac{\Sigma (x - \bar{x})^2}{n} = \frac{10}{5} = \underline{2}$$

STEP 4: Find the standard deviation (σ).

$$\sigma = \sqrt{\sigma^2} = \sqrt{2} = \underline{1.41}$$

For Machine A:
Mean = 100
Variance = 2
Standard Deviation = 1.41

PART B : MACHINE B

STEP 1: Find the mean $(\bar{x})$.

$$\bar{x} = \frac{95 + 100 + 105 + 90 + 110}{5}$$

$$= \frac{500}{5} = \underline{100}$$

STEP 2: Prepare the table.

x	$x - \bar{x}$ $(x - \bar{x} = 100)$	$(x - \bar{x})^2$
95	95 - 100 = -5	25
100	100 - 100 = 0	0
105	105 - 100 = 5	25
90	90 - 100 = -10	100
110	110 - 100 = 10	100
$\Sigma x = 500$		$\Sigma (x - \bar{x})^2 = 250$

STEP 3: Find the variance (σ^2).

$$\sigma^2 = \frac{\Sigma (x - \bar{x})^2}{n} = \frac{250}{5} = \underline{50}$$

STEP 4: Find the standard deviation (σ).

$$\sigma = \sqrt{\sigma^2} = \sqrt{50} = \underline{7.07}$$

For Machine B:
Mean = 100
Variance = 50
Standard Deviation = 7.07

FINAL ANALYSIS:

(i) **Better Performance:** Both machines have the same mean output (100 units).
So, both machines have the **same performance.**

(ii) **More Consistent:** The machine with smaller standard deviation is more consistent.
Machine A has SD = 1.41 which is smaller than Machine B SD = 7.07.
So, **Machine A is more consistent.**

FINAL ANSWER: (i) Both machines have the same performance. (ii) Machine A is more consistent.

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10th MATHEMATICS (NEW BOOK)

EXERCISE 11.2

9. The monthly sales (rupees in lacs) for two salespersons over 6 months are:

Person A : 5.5, 5.7, 5.4, 5.6, 5.8, 5.6

Person B : 5.5, 6.5, 4.5, 6.0, 5.0, 6.0

Compare their performance and consistency.

FORMULAS USED :

$$\text{Mean } (\bar{x}) = \frac{\Sigma x}{n}$$

$$\text{Variance } (\sigma^2) = \frac{\Sigma (x - \bar{x})^2}{n}$$

$$\text{Standard Deviation } (\sigma) = \sqrt{\sigma^2}$$

PART A : PERSON A

STEP 1: Find the mean $(\bar{x})$.

$$\bar{x} = \frac{5.5 + 5.7 + 5.4 + 5.6 + 5.8 + 5.6}{6}$$

$$= \frac{33.6}{6} = \boxed{5.6}$$

STEP 2: Prepare the table.

x	$x - 5.6$	$(x - 5.6)^2$
5.5	$5.5 - 5.6 = -0.1$	0.01
5.7	$5.7 - 5.6 = 0.1$	0.01
5.4	$5.4 - 5.6 = -0.2$	0.04
5.6	$5.6 - 5.6 = 0$	0
5.8	$5.8 - 5.6 = 0.2$	0.04
5.6	$5.6 - 5.6 = 0$	0
$\Sigma x = 33.6$		$\Sigma (x - \bar{x})^2 = 0.10$

STEP 3: Find the variance (σ^2).

$$\sigma^2 = \frac{\Sigma (x - \bar{x})^2}{n} = \frac{0.10}{6}$$

$$= \boxed{0.0167}$$

STEP 4: Find the standard deviation (σ).

$$\sigma = \sqrt{\sigma^2} = \sqrt{0.0167}$$

$$= 0.1291 \approx \boxed{0.13}$$

For Person A :

Mean = 5.6

Variance = 0.0167

Standard Deviation = 0.13

PART B : PERSON B

STEP 1: Find the mean $(\bar{x})$.

$$\bar{x} = \frac{95 + 100 + 105 + 90 + 110}{5}$$

$$= \frac{500}{5} = \boxed{100}$$

STEP 2: Prepare the table.

x	$x - 5.58$	$(x - 5.58)^2$
95	$95 - 100 = -5$	25
100	$100 - 100 = 0$	0
105	$105 - 100 = 5$	25
90	$90 - 100 = -10$	100
110	$110 - 100 = 10$	100
$\Sigma x = 500$		$\Sigma (x - \bar{x})^2 = 250$

STEP 3: Find the variance (σ^2).

$$\sigma^2 = \frac{\Sigma (x - \bar{x})^2}{n} = \frac{250}{5} = \underline{50}$$

STEP 4: Find the standard deviation (σ).

$$\sigma = \sqrt{\sigma^2} = \sqrt{50} = \boxed{7.07}$$

For Person B :

Mean = 5.58

Variance = 0.4514

Standard Deviation = 0.67

FINAL ANALYSIS :

(i) **Better Performance:** Person A has mean = 5.6; Person B has mean = 5.58 $\Rightarrow$ Person A has slightly higher mean. So, Person A has better performance.

(ii) **More Consistent :** Person A has SD = 0.13; Person B has SD = 0.67 $\Rightarrow$ Smaller standard deviation means more consistency. So, Person A is more consistent.

FINAL ANSWER : (ii) Person A has better performance. (ii) Person A is more consistent.

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10th MATHEMATICS (NEW BOOK)

EXERCISE 11.2

10. The table given below shows the daily wages of workers in a textile mill, grouped into six income brackets.
Calculate the mean, variance and standard deviation of the wages.

Daily Wage (Rs.)	800 – 1000	1000 – 1200	1200 – 1400	1400 – 1600	1600 – 1800	1800 – 2000
Frequency (f)	2	4	6	8	2	1

FORMULAS USED :

$$\text{Mean } (\bar{x}) = \frac{\Sigma fx}{n}$$

$$\text{Variance } (\sigma^2) = \frac{\Sigma (x - \bar{x})^2}{n}$$

$$\text{Standard Deviation } (\sigma) = \sqrt{\sigma^2}$$

STEP 1 : Find the class marks (midpoints) and prepare the table.

Class (Daily Wag e in Rs.)	Frequency (f)	Class Mark (Midpoint) (x)	fx	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
800 – 1000	2	$\frac{800 + 1000}{2} = 900$	$2 \times 900 = 1800$	$900 - \bar{x}$	$(900 - \bar{x})^2$	$2\ (900 - \bar{x})^2$
1000 – 1200	4	$\frac{1000 + 1200}{2} = 1100$	$4 \times 1100 = 4400$	$1100 - \bar{x}$	$(1100 - \bar{x})^2$	$4\ (1100 - \bar{x})^2$
1200 – 1400	6	$\frac{1200 + 1400}{2} = 1300$	$6 \times 1300 = 7800$	$1300 - \bar{x}$	$(1300 - \bar{x})^2$	$6\ (1300 - \bar{x})^2$
1400 – 1600	8	$\frac{1400 + 1600}{2} = 1500$	$8 \times 1500 = 12000$	$1500 - \bar{x}$	$(1500 - \bar{x})^2$	$8\ (1500 - \bar{x})^2$
1600 – 1800	2	$\frac{1600 + 1800}{2} = 1700$	$2 \times 1700 = 3400$	$1700 - \bar{x}$	$(1700 - \bar{x})^2$	$2\ (1700 - \bar{x})^2$
1800 – 2000	1	$\frac{1800 + 2000}{2} = 1900$	$1 \times 1900 = 1900$	$1900 - \bar{x}$	$(1900 - \bar{x})^2$	$1\ (1900 - \bar{x})^2$
	$\Sigma f = 23$		$\Sigma fx = 31300$		$\Sigma f(x - \bar{x})^2 = 1394792.69$	

STEP 2 : Find the mean ($\bar{x}$).

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{31300}{23} = \boxed{1360.87}$$

Mean daily wage ($\bar{x}$) = Rs. 1360.87

STEP 3 : Find the variance (σ^2).

$$\sigma^2 = \frac{\Sigma (x - \bar{x})^2}{\Sigma f} = \frac{60642.72}{23} = \boxed{60642.72}$$

x	f	$x - \bar{x}$ ($\bar{x} = 1360.87$)	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
900	2	900 – 1360.87 = -460.87	212401.97	424803.94
1100	4	1100 – 1360.87 = -260.87	68051.69	272206.76
1300	6	1300 – 1360.87 = -60.87	3705.57	22233.42
1500	8	1500 – 1360.87 = 139.13	19363.97	154911.76
1700	2	1700 – 1360.87 = 339.13	115001.17	230002.34
1900	1	1900 – 1360.87 = 539.13	290663.32	290663.32
$\Sigma f = 23$			$\Sigma f(x - \bar{x})^2 = 1394792.69$	

STEP 4 : Find the standard deviation (σ).

$$\sigma = \sqrt{\sigma^2} = \sqrt{60642.72} = \boxed{246.26}$$

Standard Deviation (σ) = Rs. 246.26

FINAL ANSWER :

Mean ($\bar{x}$) = Rs. 1360.87
Variance (σ^2) = 60642.72
Standard Deviation (σ) = 246.26

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10th MATHEMATICS (NEW BOOK)

EXERCISE 11.2

12. Unemployment rates (%) in five provinces are 5.2, 6.0, 4.8, 5.5, 6.2. Calculate standard deviation and describe is there balanced unemployment rate?

FORMULAS USED :

Mean $(\bar{x}) = \dfrac{\Sigma x}{n}$

Variance $(\sigma^2) = \dfrac{\Sigma (x - \bar{x})^2}{n}$

Standard Deviation $(\sigma) = \sqrt{\sigma^2}$

PART A : PERSON A

STEP 1: Find the mean $(\bar{x})$.

$$\bar{x} = \frac{5.5 + 5.7 + 5.4 + 5.6 + 5.8 + 5.6}{6}$$

$$= \frac{27.7}{5} = \boxed{5.6}$$

STEP 2: Prepare the table.

x	$x - 5.6$	$(x - 5.6)^2$
5.5	5.5 - 5.6 = -0.1	0.01
5.7	5.7 - 5.6 = 0.1	0.01
5.4	5.4 - 5.6 = -0.2	0.04
5.6	5.6 - 5.6 = 0	0
5.8	5.8 - 5.6 = 0.2	0.04
5.6	5.6 - 5.6 = 0	0
$\Sigma x = 33.6$		$\Sigma (x - \bar{x})^2 = 0.10$

STEP 3: Find the variance (σ^2).

$$\sigma^2 = \frac{\Sigma (x - \bar{x})^2}{n} = \frac{0.10}{6}$$

$$= \boxed{0.0167}$$

STEP 4: Find the standard deviation (σ).

$$\sigma = \sqrt{\sigma^2} = \sqrt{0.0167}$$

$$= 0.1291 \approx \boxed{0.13}$$

For Person A :

Mean = 5.6

Variance = 0.0167

Standard Deviation = 0.13

PART B : PERSON B

STEP 1: Find the mean $(\bar{x})$.

$$\bar{x} = \frac{95 + 100 + 105 + 90 + 110}{5}$$

$$= \frac{500}{5} = \boxed{100}$$

STEP 2: Prepare the table.

x	$x - 5.58$	$(x - 5.58)^2$
95	95 - 100 = -5	25
100	100 - 100 = 0	0
105	105 - 100 = 5	25
90	90 - 100 = -10	100
110	110 - 100 = 10	100
$\Sigma x = 500$		$\Sigma (x - \bar{x})^2 = 250$

STEP 3: Find the variance (σ^2).

$$\sigma^2 = \frac{\Sigma (x - \bar{x})^2}{n} = \frac{250}{5} = \underline{50}$$

STEP 4: Find the standard deviation (σ).

$$\sigma = \sqrt{\sigma^2} = \sqrt{50} = \boxed{7.07}$$

For Person B :

Mean = 5.58

Variance = 0.4514

Standard Deviation = 0.67

FINAL ANALYSIS :

(i) **Better Performance:** Person A has mean = 5.6; Person B has mean = 5.58 $\Rightarrow$ Person A has slightly higher mean. So, Person A has better performance.

(ii) **More Consistent :** Person A has SD = 0.13; Person B has SD = 0.67 $\Rightarrow$ Smaller standard deviation means more consistency. So, Person A is more consistent.

FINAL ANSWER : (i) Person A has better performance. (ii) Person A is more consistent.

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REVIEW EXERCISE 11.1

2. The following results for the long jump were recorded:

Distance (in cm)	170–180	180–190	190–200	200–210	210–230
f (Frequency)	2	6	9	12	3

Construct the cumulative frequency polygon and locate median, Q_1, Q_3, D_7, P_{90} and interquartile range on it.

Note: Read the values directly from the box-and-whisker plot.

★ STEP 1: Less-than type cumulative frequency table.

Class Interval (Distance in cm)	Frequency (f)	Cumulative Frequency (C.F.)
170 – 180	2	2
180 – 190	6	2 + 6 = 8
190 – 200	9	8 + 9 = 17
200 – 210	12	17 + 12 = 29
210 – 220	8	29 + 8 = 37
220 – 230	3	37 + 3 = 40

Total Frequency (N) = 40

★ STEP 2: Find the positions of required values.

- Minimum value = $\frac{N}{2} = \frac{40}{2} = 20$
- Q_1 (First quartile) = $\frac{N}{4} = \frac{40}{4} = 10$
- Median value = Line (or dot) inside the box = 25
- Q_3 (Third quartile) = $\frac{3N}{4} = \frac{3 \times 40}{4} = 36$
- Maximum value = Right end of right whisker = 46
- 90th Percentile (P_{90}) position = $\frac{90N}{100} = \frac{90 \times 40}{100} = 36$

★ STEP 3: Read the values from the cumulative frequency polygon.

Median (at 20) ⇒ 202.5 cm

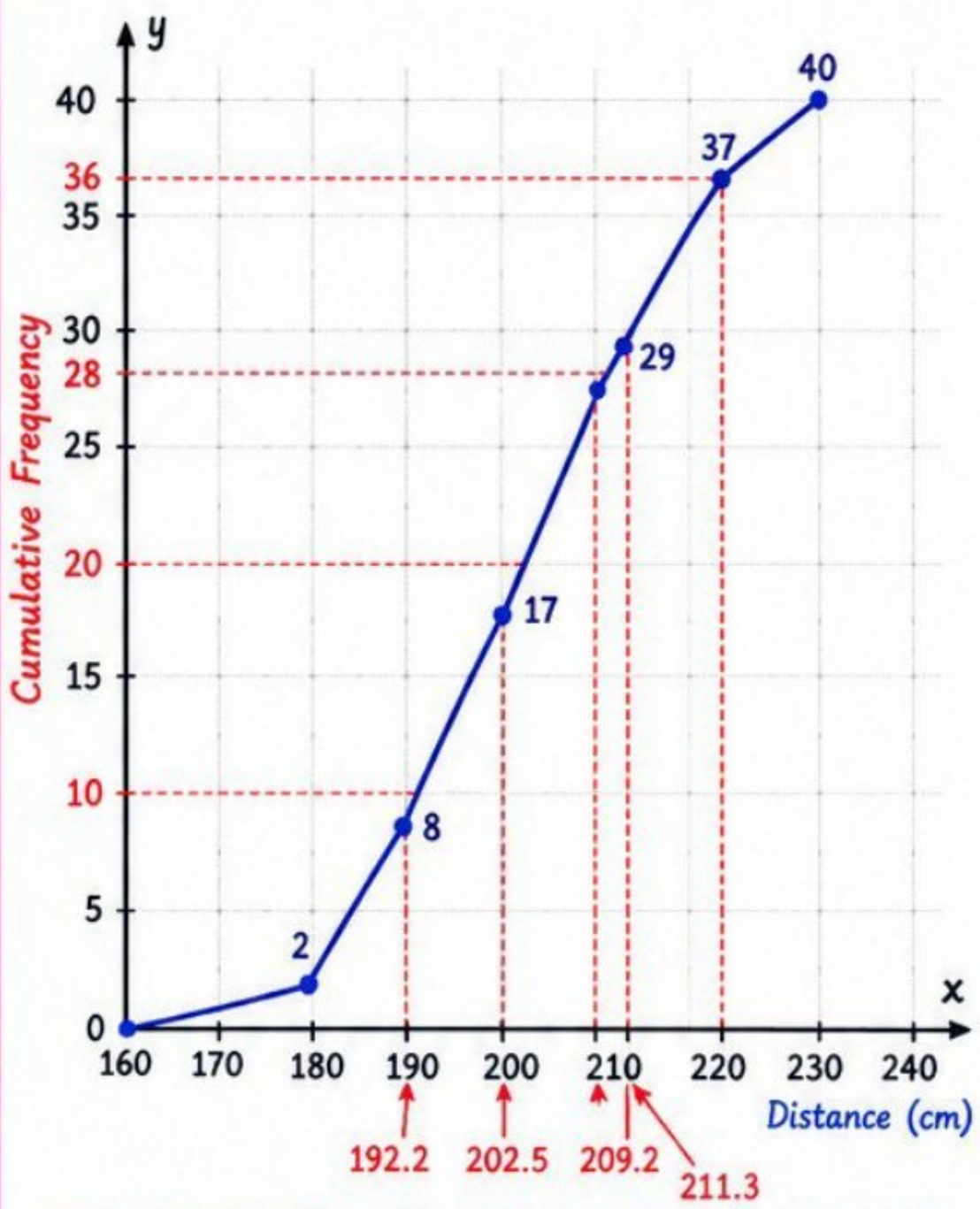
First Quartile Q_1 (at 10) ⇒ 192.2 cm

Third Quartile Q_3 (at 30) ⇒ 211.3 cm

Seventh Decile D_7 (at 28) ⇒ 209.2 cm

90th Percentile P_{90} (at 36) ⇒ 218.8 cm

★ STEP 4: Cumulative Frequency Polygon (Less-than type).



STEP 5: Find Interquartile Range (IQR).

$IQR = Q_3 - Q_1$

$= 211.3 - 192.2$

$= \boxed{19.1 \text{ cm}}$

FINAL ANSWERS:

- ★ Median (M) = 202.5 cm
- ★ First Quartile (Q_1) = 192.2 cm
- ★ Third Quartile (Q_3) = 211.3 cm
- ★ Seventh Decile (D_7) = 209.2 cm
- ★ 90th Percentile (P_{90}) = 218.8 cm
- ★ Interquartile Range (IQR) = 19.1 cm

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REVIEW EXERCISE 11.1

3. The summary statistics for a data set is given below. Show it with a box-and-whisker plot.

Min. Value	Max. Value	Q_1	Median	Q_3
5	70	12.6	43	55.6

Note: Read the values directly from the box-and-whisker plot.

SOLUTION :

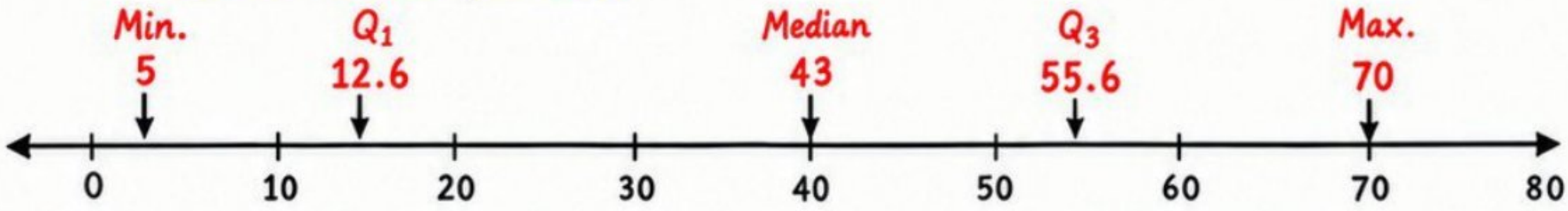
A box-and-whisker plot is drawn using the five-number summary.

STEP 1 : Write the five-number summary.

⇒ Minimum Value = 5
⇒ First Quartile (Q_1) = 12.6
⇒ Median = 43
⇒ Third Quartile (Q_3) = 55.6
⇒ Maximum Value = 70

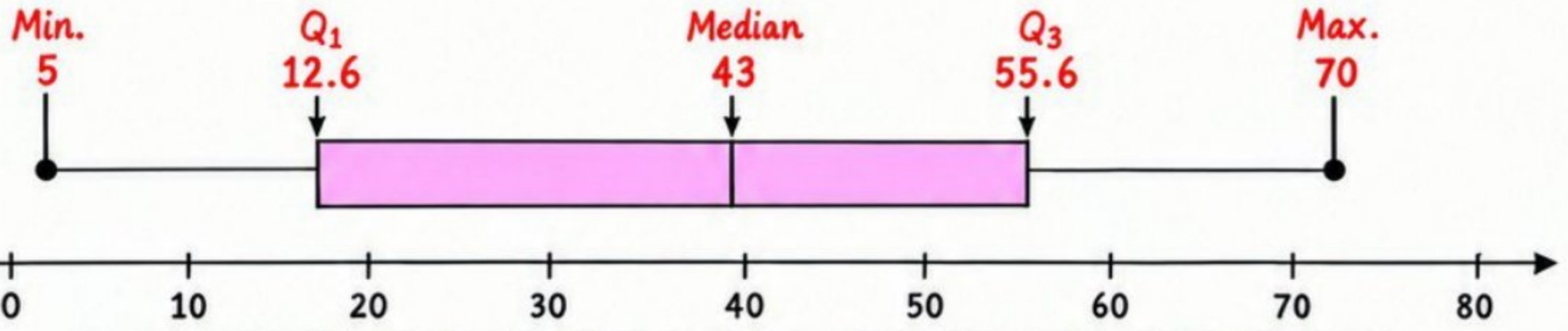
Five-number Summary consists of the smallest value, first quartile, median, third quartile and largest value.

STEP 2 : Draw a horizontal number line and mark the five values in their correct positions.



STEP 3 : Draw the box from Q_1 to Q_3 and the median as a vertical line inside the box.

STEP 4 : Draw the whiskers from the minimum value to Q_1 and from Q_3 to maximum value.



Thus, the box-and-whisker plot for the given data set is shown above.

FIVE-NUMBER SUMMARY :

Min. = 5 | Q_1 = 12.6 | Median = 43 | Q_3 = 55.6 | Max. = 70

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REVIEW EXERCISE 11

4. The following table shows the weekly hours spent exercising and resting heart rate (beats per minute) for ten individuals:

Exercise Hours (x)	0	1	2	3	4	5	6	7	8	9
Heart Rate (y)	90	85	83	80	76	74	72	70	68	65

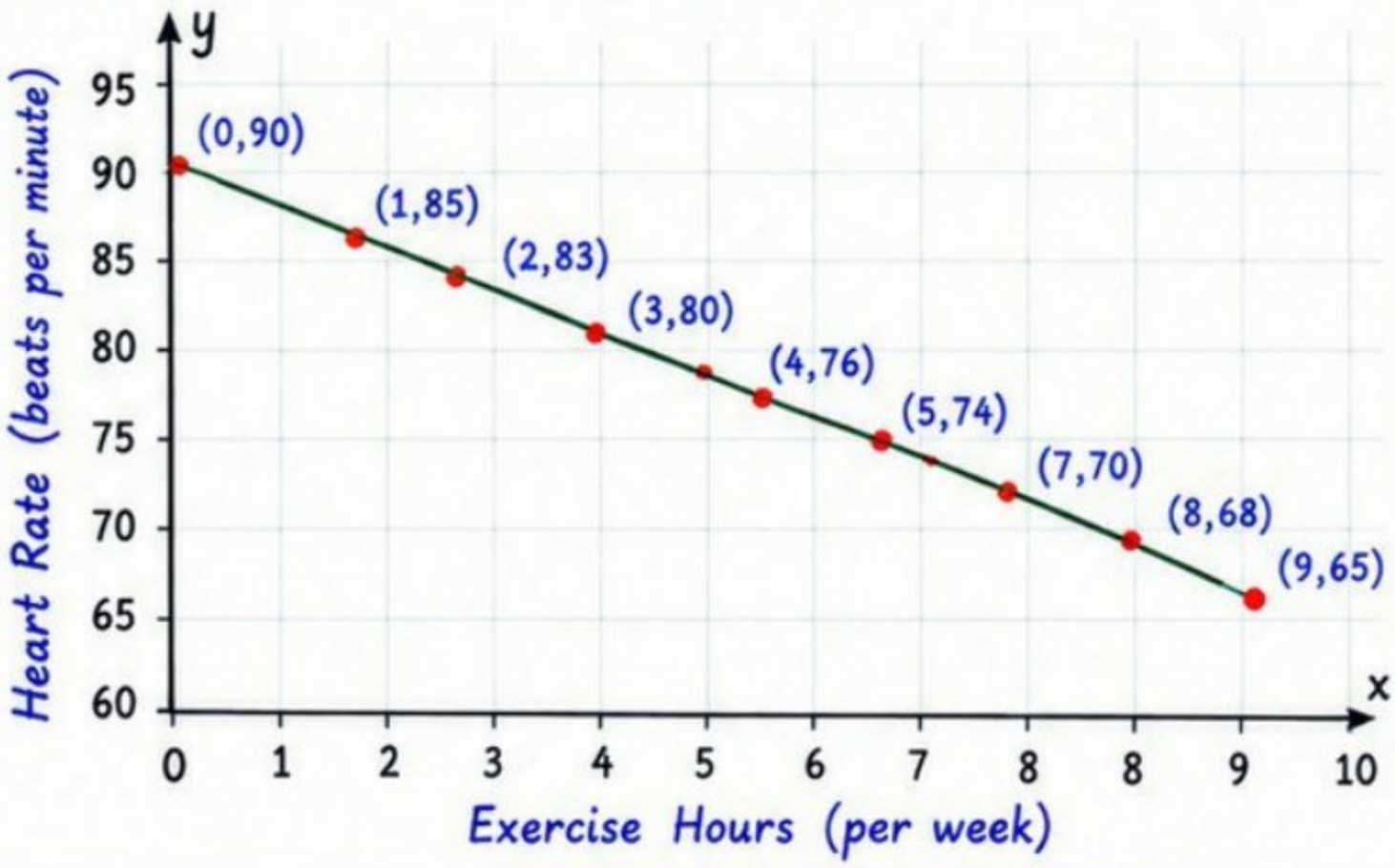
(i) Plot the data on a scatter diagram and draw a line of best fit.

(ii) State the type of correlation observed.

(iii) Predict the heart rate of Sakeena who exercises for 6.5 hours per week.

SOLUTION :

(i) Scatter Diagram with Line of Best Fit



(ii) Type of Correlation

As the number of exercise hours increases, the heart rate decreases.

The points fall close to a downward sloping straight line.

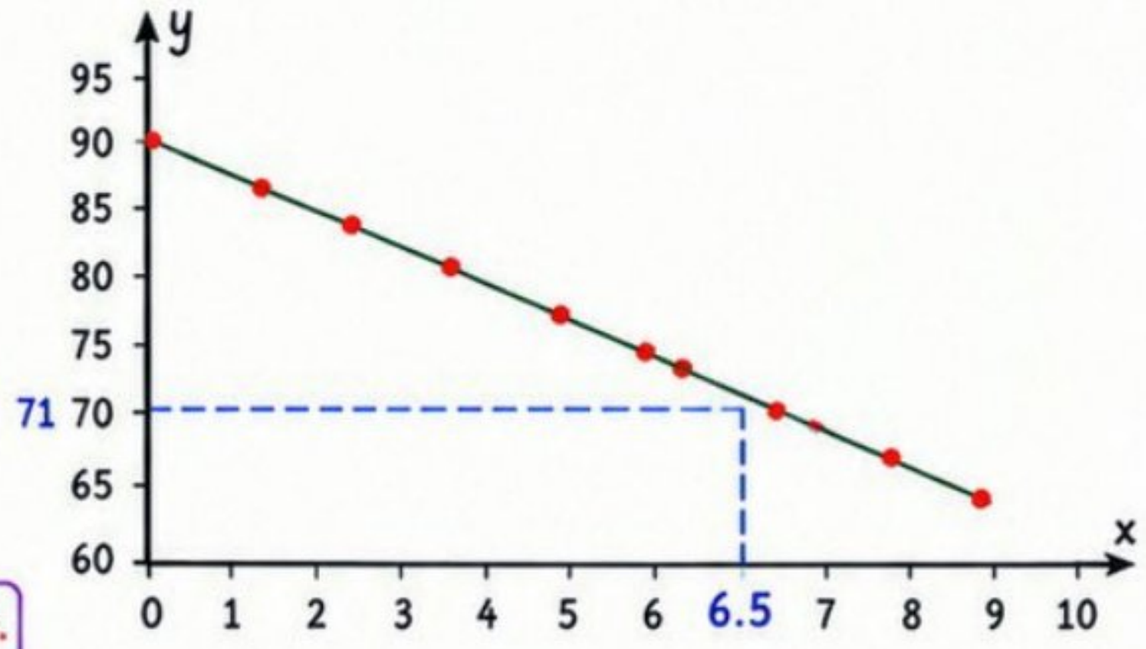
Therefore, the correlation is **Negative**.

(iii) Prediction of Heart Rate for 6.5 Hours

To find the predicted heart rate at $x = 6.5$ hours, we will use the line of best fit.

From the graph, draw a vertical line from $x = 6.5$ to the line of best fit and then a horizontal line to the y-axis.

Reading from the y-axis, we get:



Predicted Heart Rate at 6.5 hours ≈ 71 beats per minute.

FIVE-NUMBER SUMMARY:

Min. = 5 | Q_1 = 12.6 | Median = 43 | Q_3 = 55.6 | Max. = 70

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REVIEW EXERCISE 11

5. Find the range for the given data.

25 , 30 , 35 , 40 , 50 , 60 , 65 , 75

SOLUTION :

STEP 1 : Arrange the data in ascending order.

25 , 30 , 35 , 40 , 50 , 60 , 65 , 75

STEP 2 : Identify the highest and the lowest values.

$\Rightarrow$ Highest value = 75

$\Rightarrow$ Lowest value = 25

Key Point

Range tells us how much the data is spread out.

STEP 3 : Apply the formula.

$$\text{Range} = \text{Highest value} - \text{Lowest value}$$

STEP 4 : Substitute the values.

$$\text{Range} = 75 - 25$$

$$\text{Range} = 50$$

FINAL ANSWER : Range = **50**

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REVIEW EXERCISE 11

6. Calculate range, variance and standard deviation for the following data set:

Class Interval	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25	25 – 30
f (Frequency)	4	12	20	24	16	4

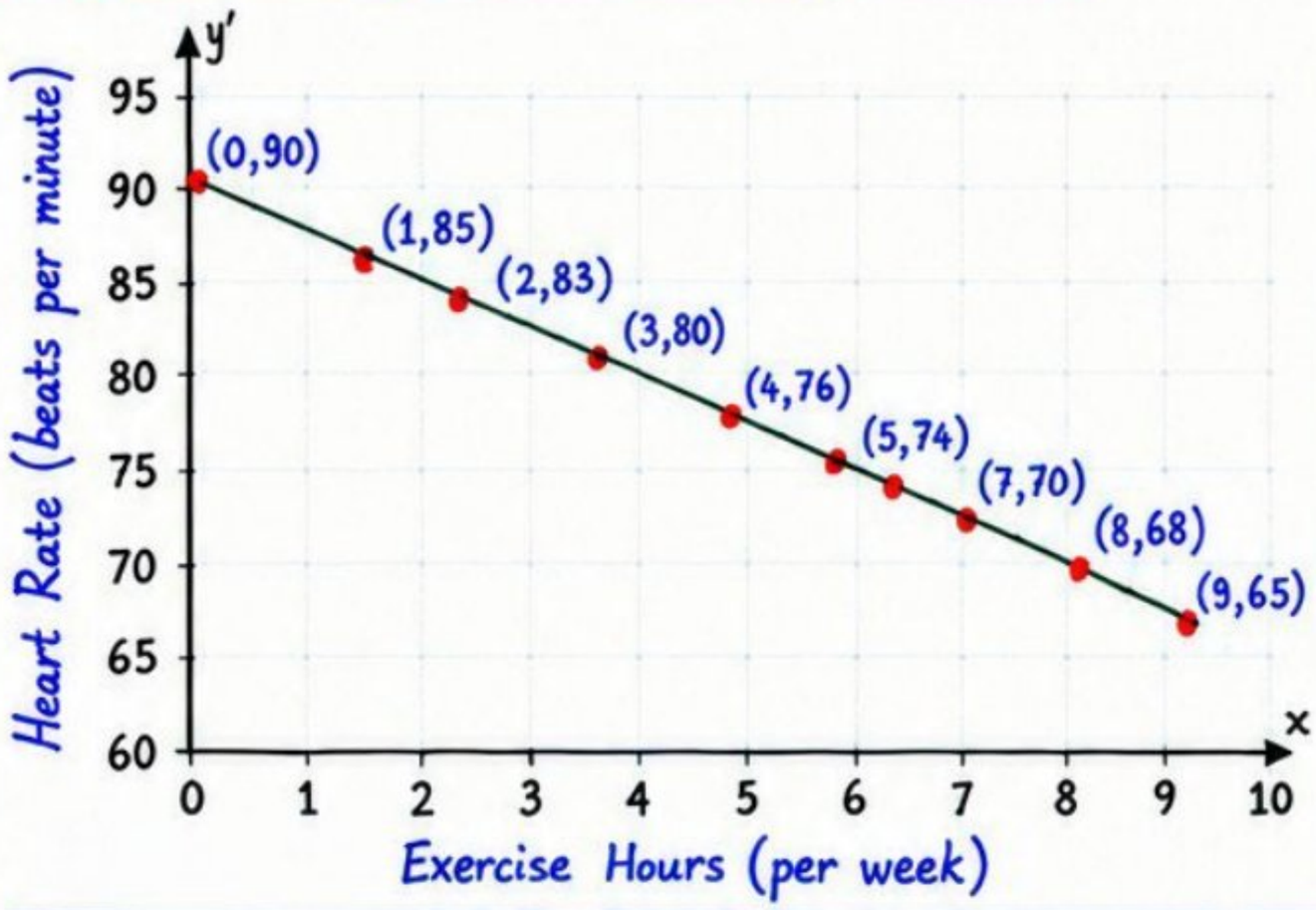
(i) Plot the data on a scatter diagram and draw a line of best fit.

(ii) State the type of correlation observed.

(iii) Predict the heart rate of Sakeena who exercises for 6.5 hours per week.

SOLUTION :

(i) Scatter Diagram with Line of Best Fit



(ii) Type of Correlation

As the number of exercise hours increases, the heart rate decreases.

The points fall close to a downward sloping straight line.

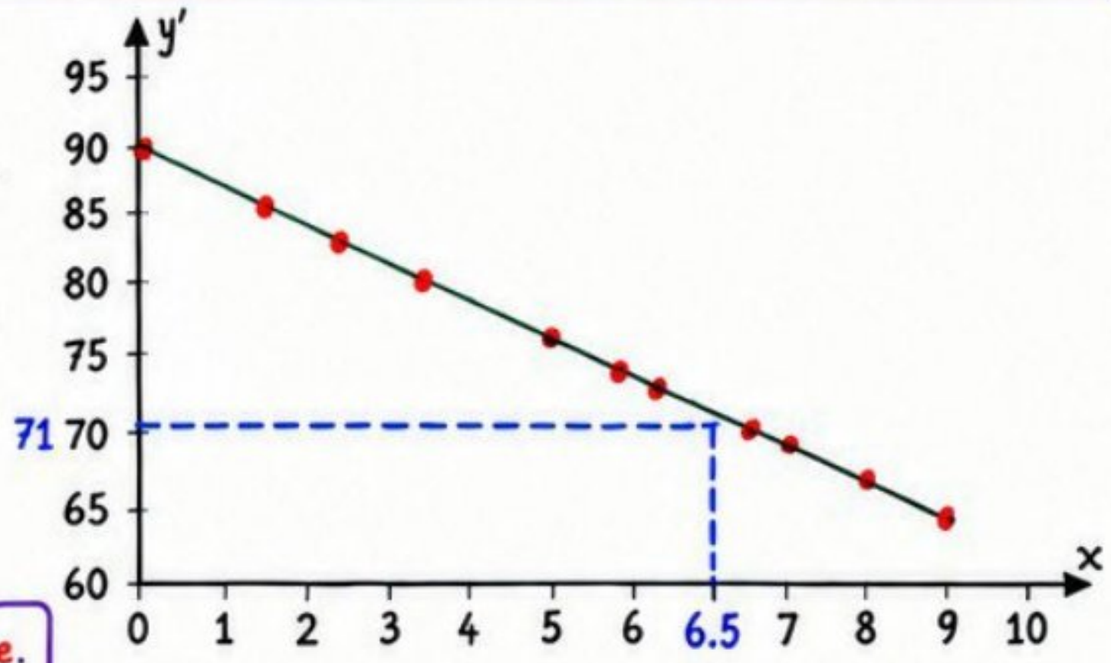
Therefore, the correlation is **Negative.**

(iii) Prediction of Heart Rate for 6.5 Hours

To find the predicted heart rate at x = 6.5 hours, we will use the line of best fit.

From the graph, draw a vertical line from x = 6.5 to the line of best fit and then a horizontal line to the y-axis.

Reading from the y-axis, we get:



Predicted Heart Rate at 6.5 hours ≈ 71 beats per minute.

FINAL ANSWERS :

(i) Scatter diagram with line of best fit is drawn above.

(ii) Type of correlation is Negative.

(iii) Predicted heart rate at 6.5 hours ≈ 71 bpm.

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REVIEW EXERCISE 11

7. The sum of 5 numbers is 45 and the sum of their squares is 421. Find the mean and standard deviation of the data.

STEP 1 : Write the given information.

Number of values, $n = 5$

Sum of the values, $\Sigma x = 45$

Sum of the squares, $\Sigma x^2 = 421$

STEP 2 : Find the mean $(\bar{x})$.

$$\bar{x} = \frac{\Sigma x}{n}$$

$$\bar{x} = \frac{45}{5}$$

$$\bar{x} = 9$$

Key Point

Mean $(\bar{x})$ is the average of all the values. We divide the sum of values by the total number of values.

STEP 3 : Find the variance (σ^2).

$$\sigma^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2$$

Substitute the values.

$$\sigma^2 = \frac{421}{5} - (9)^2$$

$$\sigma^2 = 84.2 - 81$$

$$\sigma^2 = 3.2$$

STEP 4 : Find the standard deviation (σ).

$$\sigma = \sqrt{\sigma^2}$$

$$\sigma = \sqrt{3.2}$$

$$\sigma \approx 1.79$$

Key Point

Standard deviation (σ) is the positive square root of variance.

STEP 5 : Find standard deviation (σ).

$$\sigma = \sqrt{\sigma^2} = \sqrt{38.5} = 6.20 \text{ (approx.)}$$

FINAL ANSWERS :

Range	=	30
Mean $(\bar{x})$	=	15.5
Variance (σ^2)	=	38.5
Standard Deviation (σ)	=	6.20 (approx.)

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REVIEW EXERCISE 11

8. The monthly household expenses (rupees in thousands) of two families for 6 months were :

Family A : 45, 47, 46, 48, 46, 47

Family B : 38, 52, 40, 50, 42, 49

Calculate the mean and standard deviation of the monthly expenses.

Which family spends more on average? Which family has more stable expenses?

SOLUTION :

FAMILY A : 45, 47, 46, 48, 46, 47

STEP 1 : Find the mean ($\bar{x}$).

$$\bar{x} = \frac{\text{Sum of values}}{n} = \frac{45+47+46+48+46+47}{6}$$

$$= \frac{279}{6} = \boxed{46.5}$$

STEP 2 : Find the variance (σ^2).

$$\sigma^2 = \frac{\Sigma(x-\bar{x})^2}{n}$$

x	$x-\bar{x}$	$(x-\bar{x})^2$
45	45 - 46.5 = -1.5	2.25
47	47 - 46.5 = 0.5	0.25
46	46 - 46.5 = -0.5	0.25
48	48 - 46.5 = 1.5	2.25
46	46 - 46.5 = -0.5	0.25
47	47 - 46.5 = 0.5	0.25
		$\Sigma(x-\bar{x})^2 = 5.50$

$$\sigma^2 = \frac{5.50}{6} \approx \boxed{0.9167}$$

STEP 3 : Find the standard deviation (σ).

$$\sigma = \sqrt{\sigma^2}$$

$$\sigma = \sqrt{0.9167} \approx \boxed{0.96}$$

Key Point

Variance (σ^2) is the average of the squares of deviations from the mean. Standard deviation (σ) is the square root of variance.

FAMILY B : 38, 52, 40, 50, 42, 49

STEP 1 : Find the mean ($\bar{x}$).

$$\bar{x} = \frac{38+52+40+50+42+49}{6} = \frac{271}{6} \approx \boxed{45.17}$$

STEP 2 : Find the variance (σ^2).

$$\sigma^2 = \frac{\Sigma(x-\bar{x})^2}{n}$$

x	$x-\bar{x}$	$(x-\bar{x})^2$
38	38 - 45.17 = -7.17	51.41
52	52 - 45.17 = 6.83	46.64
40	40 - 45.17 = -5.17	26.73
50	50 - 45.17 = 4.83	23.33
42	42 - 45.17 = -3.17	10.05
49	49 - 45.17 = 3.83	14.66
		$\Sigma(x-\bar{x})^2 = 172.82$

$$\sigma^2 = \frac{172.82}{6} \approx \boxed{28.81}$$

STEP 3 : Find the standard deviation (σ).

$$\sigma = \sqrt{\sigma^2}$$

$$\sigma = \sqrt{28.81} \approx \boxed{5.37}$$

Key Point

Standard deviation (σ) is the positive square root of variance.

COMPARISON AND CONCLUSION :

- Mean of Family A = 46.5
- Mean of Family B ≈ 45.17

⇒ Family A spends more on average.

- Standard Deviation of Family A ≈ 0.96
- Standard Deviation of Family B ≈ 5.37

FINAL ANSWERS :

Family A : Mean = 46.5 , Standard Deviation ≈ 0.96

- Higher average spending : Family A
- More stable expenses : Family A

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